

Diffusion-Driven Graph Optimization Framework for Bundle Recommendation

Anonymous submission

Abstract

Existing bundle recommendation methods typically use multi-view information from user–bundle, user–item, and bundle–item graphs to accurately model users’ bundle preferences. However, such multi-view methods still face two key challenges when learning from these graphs: (1) unreliable user–bundle interactions may propagate misleading signals and impair preference learning; (2) the sparsity of user–bundle implicit feedback limits the available supervision. We therefore propose a diffusion-driven graph optimization framework for bundle recommendation (DiGO). First, we aggregate pretrained representations of the bundles in each user’s interaction history and condition a diffusion process on the pretrained user representation to model the user’s overall bundle preference. We then use this preference representation to score observed user–bundle interactions, retain more reliable user–bundle interactions, and reconstruct the user–bundle graph for subsequent multi-view representation learning. Second, to address the sparsity of user–bundle implicit feedback, we use representations learned from the reconstructed user–bundle graph as anchors and align user–item and bundle–item representations through cross-view contrastive learning before fusing the three views. This allows the auxiliary relations to supplement the limited supervision from user–bundle feedback. Experiments on three public datasets show that our framework outperforms state-of-the-art methods on all evaluation metrics, achieving a maximum relative improvement of 10% over the strongest baseline.

1 Introduction

Bundle recommendation aims to identify bundles that match users’ interests from a set of predefined item collections. Existing methods typically model user–bundle, user–item, and bundle–item relations jointly and learn complementary user and bundle representations through graph neural networks, multi-view fusion, and contrastive learning (????). Among these relations, user–item and bundle–item relations provide auxiliary information, whereas the user–bundle graph directly carries users’ implicit feedback on bundles. Its structural and supervisory quality therefore affects subsequent

multi-view learning. Despite the progress of existing methods, multi-view bundle recommendation still faces two key challenges.

Challenge 1: Unreliable interactions in the user–bundle graph. User–bundle relations are typically constructed from implicit feedback and may contain interactions that do not reliably reflect users’ preferences. For example, accidental clicks, exploratory choices, and interactions caused by short-term interests or exposure bias may fail to consistently represent users’ actual bundle preferences. Existing graph-based methods propagate information directly along observed edges, allowing unreliable interactions to participate in representation updates and introduce misleading signals. Therefore, identifying more reliable connections from users’ interaction histories and constructing a high-quality user–bundle graph remain key challenges.

Challenge 2: Sparse user–bundle implicit feedback. User–bundle feedback is substantially sparser than user–item interactions, providing limited bundle-level supervision. Although user–item and bundle–item relations offer rich auxiliary signals, final-stage fusion alone may not fully align them with user–bundle preference learning. Therefore, effectively leveraging auxiliary views while preserving consistency with direct bundle preferences remains challenging.

To address these challenges, we propose DiGO, a diffusion-driven graph optimization framework for bundle recommendation. DiGO first aggregates pretrained representations of each user’s historical bundles and conditions a diffusion process on the pretrained user representation to model the user’s overall bundle preference. The resulting preference representation is then used to score observed user–bundle interactions, retain more reliable connections, and reconstruct the graph without introducing new edges. Finally, DiGO uses representations from the reconstructed user–bundle graph as anchors to align the user–item and bundle–item views before fusion, thereby supplementing sparse supervision. Figure 1 illustrates the overall framework.

The main contributions of this work are summarized as follows:

- We propose DiGO, a diffusion-driven graph optimization framework that models users’ overall bundle preferences through conditional diffusion and uses them to re-evaluate observed interactions and reconstruct the user–bundle graph.

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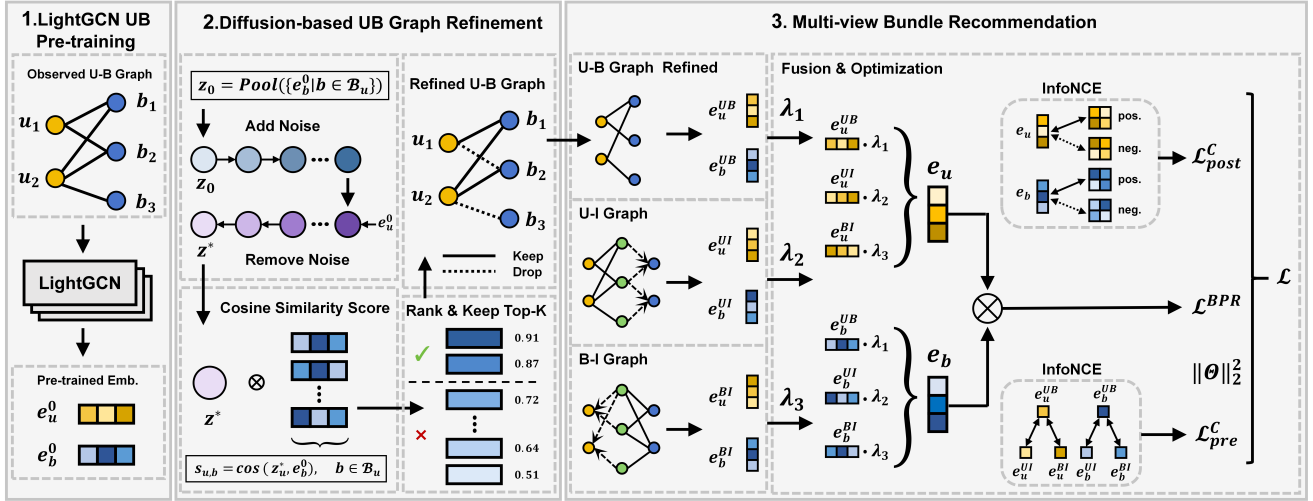


Figure 1: Overall framework of DiGO, comprising (1) user–bundle pretraining, (2) diffusion-driven graph optimization for reconstructing the observed user–bundle graph, and (3) multi-view recommendation using the reconstructed user–bundle view as the anchor for pre-fusion cross-view contrastive learning.

- We use representations learned from the reconstructed user–bundle graph as anchors to align the two auxiliary views before view fusion, thereby supplementing sparse user–bundle supervision.
- Experiments on three public datasets show that DiGO outperforms existing methods across all evaluation metrics, achieving a maximum relative improvement of 10%. Ablation and further analyses also validate the effectiveness of the two key designs.

2 Related Work

2.1 Bundle Recommendation

Early bundle recommendation methods primarily relied on user–bundle interactions and employed collaborative filtering, matrix factorization, or neural models to learn user and bundle representations (??). Subsequent graph-based methods further modeled user–bundle, user–item, and bundle–item relations to capture high-order collaborative information at both the bundle and item levels (??).

Recent studies have increasingly focused on multi-view representation learning and cross-view collaboration. For example, MIDGN introduces intent disentanglement, CrossCBR employs cross-view contrastive learning, BundleGT uses graph Transformers to model structural information, and MultiCBR learns unified representations through three-view fusion and contrastive learning (????). These methods mainly improve the encoding, fusion, or alignment of multiple views, while typically using the observed user–bundle graph directly. In contrast, DiGO re-evaluates observed user–bundle interactions before multi-view learning and reconstructs the user–bundle graph accordingly.

2.2 Diffusion Models for Recommendation

Diffusion models exhibit strong distribution modeling capabilities by progressively perturbing data and learning the cor-

responding reverse process. In recent years, they have been applied to collaborative filtering, sequential recommendation, and knowledge graph recommendation to model user preferences, recover interaction signals, or enhance recommendation representations (??).

Diffusion models have also been introduced into bundle recommendation. RDiffBR addresses dynamic bundle–item relations by applying residual diffusion to item-level bundle representations, enabling the model to adapt to changes in bundle composition (?). DCBR applies conditional diffusion to user–bundle interaction data and combines it with multi-view contrastive learning (?). Unlike these methods, DiGO aggregates the pretrained representations of bundles in each user’s interaction history and conditions the diffusion process on the pretrained user representation to model the user’s overall bundle preference. The resulting preference representation is then used to evaluate observed user–bundle interactions and reconstruct the user–bundle graph for subsequent multi-view representation learning.

3 Methodology

3.1 Problem Formulation

Let $\mathcal{U} = \{u_1, u_2, \dots, u_M\}$, $\mathcal{B} = \{b_1, b_2, \dots, b_N\}$, and $\mathcal{I} = \{i_1, i_2, \dots, i_O\}$ denote the sets of users, bundles, and items, respectively, where M , N , and O are the corresponding set sizes. We use a user–bundle interaction matrix $\mathbf{X} \in \{0, 1\}^{M \times N}$, a user–item interaction matrix $\mathbf{Y} \in \{0, 1\}^{M \times O}$, and a bundle–item affiliation matrix $\mathbf{Z} \in \{0, 1\}^{N \times O}$. Specifically, $x_{ub} = 1$ and $y_{ui} = 1$ indicate that user u has interacted with bundle b and item i , respectively, while $z_{bi} = 1$ indicates that item i belongs to bundle b .

Given $\mathbf{X}$, $\mathbf{Y}$, and $\mathbf{Z}$, the goal of bundle recommendation is to predict the preference score $y_{u,b}^*$ for each unobserved user–bundle pair.

3.2 Framework Overview

As illustrated in Figure 1, DiGO first models each user’s overall bundle preference using pretrained representations and reconstructs the original user–bundle matrix $\mathbf{X}$ as $\tilde{\mathbf{X}}$. It then learns user and bundle representations from three views based on $\tilde{\mathbf{X}}$, $\mathbf{Y}$, and $\mathbf{Z}$. Before fusing the three views, DiGO performs cross-view alignment using the user–bundle view as the anchor and finally predicts user preferences from the fused representations.

3.3 Diffusion-Driven Graph Optimization

Preference Representation Construction We first pre-train LightGCN (?) on the original user–bundle interactions to obtain pretrained representations $\bar{\mathbf{e}}_u$ and $\bar{\mathbf{e}}_b$ for user u and bundle b , respectively. The pretraining model is optimized with the BPR objective, and the resulting representations remain fixed during subsequent graph optimization.

Let $\mathcal{N}_u^{UB} = \{b \in \mathcal{B} \mid x_{ub} = 1\}$ denote the set of bundles historically interacted with by user u . We construct the historical bundle representation through mean pooling:

$$\mathbf{p}_u^0 = \frac{1}{|\mathcal{N}_u^{UB}|} \sum_{b \in \mathcal{N}_u^{UB}} \bar{\mathbf{e}}_b, \quad (1)$$

where $\mathbf{p}_u^0$ summarizes the overall characteristics of the bundles in the user’s interaction history and serves as the modeling target of the subsequent diffusion process.

Conditional Diffusion Given the number of diffusion steps T and a noise schedule $\{\beta_t\}_{t=1}^T$, we define $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$. At timestep t , the forward perturbation of $\mathbf{p}_u^0$ is defined as

$$\mathbf{p}_u^t = \sqrt{\alpha_t} \mathbf{p}_u^0 + \sqrt{1 - \alpha_t} \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (2)$$

We use the pretrained user representation $\bar{\mathbf{e}}_u$ as a personalized condition and employ a conditional denoising network to predict the unperturbed bundle preference representation:

$$\hat{\mathbf{p}}_u^0 = f_\theta(\mathbf{p}_u^t, \bar{\mathbf{e}}_u, t), \quad (3)$$

where f_θ is an MLP-based conditional denoising network that integrates the perturbed representation, the user condition, and the timestep embedding.

During training, we uniformly sample t from $\{1, \dots, T\}$ and draw Gaussian noise ϵ to construct $\mathbf{p}_u^t$. The diffusion module is optimized by reconstructing the unperturbed historical bundle representation:

$$L^D = \mathbb{E}_{u \sim \mathcal{U}^+, t \sim \text{Unif}\{1, \dots, T\}, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \left[\|\hat{\mathbf{p}}_u^0 - \mathbf{p}_u^0\|_2^2 \right], \quad (4)$$

where $\mathcal{U}^+ = \{u \in \mathcal{U} \mid |\mathcal{N}_u^{UB}| > 0\}$. This objective encourages the conditional denoising network to recover the overall representation of the bundles in the user’s interaction history.

Interaction Scoring and Graph Reconstruction During graph reconstruction, we employ a deterministic iterative denoising procedure. The historical bundle representation is used as the initial state, and the conditional denoising network is repeatedly applied in reverse timestep order:

$$\begin{aligned} \tilde{\mathbf{p}}_u^T &= \mathbf{p}_u^0, \\ \tilde{\mathbf{p}}_u^{t-1} &= f_\theta(\tilde{\mathbf{p}}_u^t, \bar{\mathbf{e}}_u, t), \quad t = T, T-1, \dots, 1. \\ \mathbf{p}_u^* &= \tilde{\mathbf{p}}_u^0, \end{aligned} \quad (5)$$

Here, $\mathbf{p}_u^*$ denotes the resulting overall bundle preference representation of user u .

We then compute preference-matching scores only for the user’s observed interactions and retain at most the top- κ bundles:

$$\begin{aligned} s_{u,b} &= \cos(\mathbf{p}_u^*, \bar{\mathbf{e}}_b), \quad b \in \mathcal{N}_u^{UB}, \\ \tilde{\mathcal{N}}_u^{UB} &= \text{TopK}_{b \in \mathcal{N}_u^{UB}}(s_{u,b}, \kappa), \\ \tilde{x}_{ub} &= \mathbb{I}[b \in \tilde{\mathcal{N}}_u^{UB}]. \end{aligned} \quad (6)$$

Here, κ is the maximum number of interactions retained for each user, and $\mathbb{I}[\cdot]$ is the indicator function. All $\tilde{x}_{ub}$ form the reconstructed user–bundle matrix $\tilde{\mathbf{X}}$. Since $\tilde{\mathcal{N}}_u^{UB} \subseteq \mathcal{N}_u^{UB}$, DiGO only re-evaluates and filters observed user–bundle interactions without introducing unobserved connections.

3.4 Multi-View Representation Learning

Based on $\tilde{\mathbf{X}}$, $\mathbf{Y}$, and $\mathbf{Z}$, DiGO learns representations from the user–bundle interaction view, user–item interaction view, and bundle–item affiliation view. The three views share the initial trainable user, bundle, and item representations while employing independent propagation paths.

Let $\text{LG}(\cdot)$ denote a LightGCN encoder with neighborhood propagation and layer-wise aggregation. The representation learning process for the three views is written as

$$\begin{aligned} (\mathbf{E}_U^{UB}, \mathbf{E}_B^{UB}) &= \text{LG}(\tilde{\mathbf{X}}), \\ (\mathbf{E}_U^{UI}, \mathbf{E}_I^{UI}) &= \text{LG}(\mathbf{Y}), \\ \mathbf{e}_b^{UI} &= \frac{1}{|\mathcal{N}_b^{BI}|} \sum_{i \in \mathcal{N}_b^{BI}} \mathbf{e}_i^{UI}, \\ (\mathbf{E}_B^{BI}, \mathbf{E}_I^{BI}) &= \text{LG}(\mathbf{Z}), \\ \mathbf{e}_u^{BI} &= \frac{1}{|\mathcal{N}_u^{UI}|} \sum_{i \in \mathcal{N}_u^{UI}} \mathbf{e}_i^{BI}. \end{aligned} \quad (7)$$

Here, $\mathcal{N}_b^{BI}$ denotes the set of items contained in bundle b , while $\mathcal{N}_u^{UI}$ denotes the set of items interacted with by user u . The UB view directly produces the user representation $\mathbf{e}_u^{UB}$ and bundle representation $\mathbf{e}_b^{UB}$. In the UI view, item representations are aggregated according to bundle–item affiliations to obtain $\mathbf{e}_b^{UI}$. In the BI view, item representations are aggregated according to user–item interactions to obtain $\mathbf{e}_u^{BI}$. This three-view organization follows the standard multi-view modeling framework for bundle recommendation (?).

Consequently, user u obtains three view-specific representations $\{\mathbf{e}_u^{UB}, \mathbf{e}_u^{UI}, \mathbf{e}_u^{BI}\}$, while bundle b obtains $\{\mathbf{e}_b^{UB}, \mathbf{e}_b^{UI}, \mathbf{e}_b^{BI}\}$.

3.5 Cross-View Contrastive Learning

To exploit the two auxiliary views for supplementing sparse user–bundle supervision, DiGO uses the representations learned from the reconstructed user–bundle view as anchors and aligns them with the user–item and bundle–item representations before fusing the three views.

For an entity set $\mathcal{S}$, the contrastive loss between an anchor view a and an auxiliary view v is defined as

$$\ell_\tau(\mathbf{E}^a, \mathbf{E}^v; \mathcal{S}) = -\frac{1}{|\mathcal{S}|} \sum_{x \in \mathcal{S}} \log \frac{\exp(\cos(\mathbf{e}_x^a, \mathbf{e}_x^v)/\tau)}{\sum_{x' \in \mathcal{S}} \exp(\cos(\mathbf{e}_x^a, \mathbf{e}_{x'}^v)/\tau)}. \quad (8)$$

Representations of the same entity across the two views form a positive pair, while representations of the remaining entities serve as in-batch negatives.

The pre-fusion cross-view contrastive loss is defined as

$$\begin{aligned} L^A = & \ell_{\tau_a}(\mathbf{E}_U^{UB}, \mathbf{E}_U^{UI}; \mathcal{S}_U) + \ell_{\tau_a}(\mathbf{E}_U^{UB}, \mathbf{E}_U^{BI}; \mathcal{S}_U) \\ & + \ell_{\tau_a}(\mathbf{E}_B^{UB}, \mathbf{E}_B^{UI}; \mathcal{S}_B) + \ell_{\tau_a}(\mathbf{E}_B^{UB}, \mathbf{E}_B^{BI}; \mathcal{S}_B), \end{aligned} \quad (9)$$

where $\mathcal{S}_U$ and $\mathcal{S}_B$ denote the sets of unique users and positive bundles in the current batch, respectively, and τ_a is the temperature parameter. DiGO only constructs UB–UI and UB–BI alignment on both the user and bundle sides, without introducing an additional UI–BI constraint.

3.6 Prediction and Model Optimization

The user and bundle representations from the three views are fused using view coefficients, and their inner product is used for preference prediction:

$$\begin{aligned} \mathbf{e}_u &= \lambda_1 \mathbf{e}_u^{UB} + \lambda_2 \mathbf{e}_u^{UI} + \lambda_3 \mathbf{e}_u^{BI}, \\ \mathbf{e}_b &= \lambda_1 \mathbf{e}_b^{UB} + \lambda_2 \mathbf{e}_b^{UI} + \lambda_3 \mathbf{e}_b^{BI}, \\ y_{u,b}^* &= \mathbf{e}_u^\top \mathbf{e}_b, \quad \lambda_1 + \lambda_2 + \lambda_3 = 1. \end{aligned} \quad (10)$$

DiGO also retains post-fusion self-supervised contrastive learning. Let $\mathbf{E}'_U, \mathbf{E}''_U$ and $\mathbf{E}'_B, \mathbf{E}''_B$ denote the user and bundle representations generated by two independent representation perturbations, respectively. The post-fusion contrastive loss is

$$L^C = \frac{1}{2} [\ell_{\tau_c}(\mathbf{E}'_U, \mathbf{E}''_U; \mathcal{S}_U) + \ell_{\tau_c}(\mathbf{E}'_B, \mathbf{E}''_B; \mathcal{S}_B)], \quad (11)$$

where τ_c is the temperature parameter for post-fusion contrastive learning.

For each training triplet (u, b, b') , where $x_{ub} = 1$ and $x_{ub'} = 0$, the recommendation loss and the overall objective of the multi-view recommendation model are defined as

$$\begin{aligned} L^{BPR} &= \sum_{(u,b,b')} -\ln \sigma(y_{u,b}^* - y_{u,b'}^*), \\ L^R &= L^{BPR} + \beta_1 L^C + \beta_2 L^A + \beta_3 \|\Theta\|_2^2, \end{aligned} \quad (12)$$

where Θ denotes the trainable parameters of the multi-view recommendation model, and β_1, β_2 , and β_3 control the contributions of the post-fusion contrastive loss, the pre-fusion cross-view contrastive loss, and the regularization term, respectively.

During training, DiGO first optimizes the diffusion module with L^D and reconstructs $\tilde{\mathbf{X}}$. It then uses $\tilde{\mathbf{X}}, \mathbf{Y}$, and $\mathbf{Z}$ as inputs and optimizes the multi-view recommendation model with L^R .

3.7 Complexity Analysis

Let d, K , and Q denote the embedding dimension, the number of graph propagation layers, and the number of recommendation mini-batches per epoch, respectively. Let $C_f = \mathcal{O}(d^2)$ denote the cost of one forward pass through the fixed-depth denoising MLP. We use $\mathcal{E}_{UB}, \mathcal{E}_{UI}$, and $\mathcal{E}_{BI}$ to denote the edge sets of the three original graphs, and $\tilde{\mathcal{E}}_{UB}$ for the reconstructed UB graph. The graph propagation in offline LightGCN pretraining requires $\mathcal{O}(K|\mathcal{E}_{UB}|d)$ per pass. During each training epoch, the time complexity of DiGO is

$$\begin{aligned} \mathcal{O}((T+1)|\mathcal{U}^+|C_f + |\mathcal{E}_{UB}|(d + \log \kappa) \\ + QKd(|\tilde{\mathcal{E}}_{UB}| + |\mathcal{E}_{UI}| + |\mathcal{E}_{BI}|) \\ + Q(B_U^2 + B_B^2)d), \end{aligned}$$

where B_U and B_B are the numbers of unique users and bundles in a mini-batch. The first two terms jointly account for diffusion learning and graph reconstruction, while the remaining terms correspond to multi-view graph propagation and contrastive learning. Since $|\tilde{\mathcal{E}}_{UB}| \leq |\mathcal{E}_{UB}|$, DiGO does not increase propagation costs through graph densification and avoids scoring all $M \times N$ user–bundle pairs. Its trainable parameter complexity is $\mathcal{O}((M+N+O)d + P_f)$, where P_f denotes the number of parameters in the denoising network.

4 Experiments

4.1 Experimental Settings

Datasets and Evaluation Metrics. We conduct experiments on three public bundle recommendation datasets: Youshu, NetEase, and iFashion, which correspond to book lists, music playlists, and fashion outfits, respectively. For a fair comparison, we adopt the same preprocessed datasets and data splits as CrossCBR and MultiCBR (??), where user–bundle interactions are divided into training, validation, and test sets with a ratio of 70%, 10%, and 20%, respectively. The dataset statistics are summarized in Table 2.

We adopt Recall@ K and NDCG@ K as evaluation metrics, where $K \in \{10, 20\}$. Following the full-ranking evaluation protocol, all candidate bundles are ranked for each user after masking the bundles observed in the training set. We select the model achieving the best Top-20 performance on the validation set and report its performance on the test set.

Baselines. We compare DiGO with seven representative bundle recommendation methods, including BGCN, MIDGN, CrossCBR, BundleGT, MultiCBR, EBRec, and EpicCBR (???????). These methods cover representative

Method	Youshu				NetEase				iFashion			
	R@10	N@10	R@20	N@20	R@10	N@10	R@20	N@20	R@10	N@10	R@20	N@20
BGCN	–	–	0.2436	0.1329	–	–	0.0626	0.0328	–	–	0.0802	0.0582
MIDGN	–	–	0.2682	0.1527	–	–	0.0678	0.0343	–	–	0.0856	0.0473
CrossCBR	–	–	0.2823	0.1670	–	–	0.0844	0.0458	–	–	0.1132	0.0872
BundleGT	–	–	<u>0.2905</u>	<u>0.1737</u>	–	–	0.0903	0.0478	–	–	0.1263	0.0969
MultiCBR	0.1995	0.1439	0.2847	0.1681	0.0578	0.0389	0.0901	0.0491	<u>0.1057</u>	<u>0.1028</u>	<u>0.1494</u>	<u>0.1203</u>
EBRec	0.1984	0.1425	0.2797	0.1659	0.0559	0.0376	0.0881	0.0477	0.0909	0.0874	0.1327	0.1044
EpicCBR	<u>0.1999</u>	<u>0.1443</u>	0.2835	0.1682	<u>0.0601</u>	<u>0.0397</u>	<u>0.0926</u>	<u>0.0503</u>	0.1049	0.1022	0.1467	0.1190
DiGO	0.2053	0.1482	0.2919	0.1759	0.0617	0.0411	0.0955	0.0511	0.1144	0.1146	0.1568	0.1315
Improv.	2.70%	2.70%	0.48%	1.27%	2.66%	3.53%	3.13%	1.59%	8.23%	11.48%	4.95%	9.31%

Table 1: Overall performance comparison on three datasets. The best and second-best results are highlighted in bold and underlined, respectively.

Dataset	#Users	#Items	#Bundles	#U-I	#U-B	#Avg. I/B
Youshu	8,039	32,770	4,771	138,515	51,377	37.03
NetEase	18,528	123,628	22,864	1,128,065	302,303	77.80
iFashion	53,897	42,563	27,694	2,290,645	1,679,708	3.86

Table 2: Statistics of the datasets.

Variant	Youshu		NetEase		iFashion	
	R@20	N@20	R@20	N@20	R@20	N@20
DiGO	0.2919	0.1759	0.0955	0.0511	0.1568	0.1315
w/o Graph Opt.	0.2920	0.1734	0.0948	0.0509	0.1500	0.1245
w/o All CL	0.2326	0.1307	0.0512	0.0260	0.0400	0.0281
w/o Pre-fusion CL	0.2901	0.1692	0.0906	0.0487	0.1558	0.1304
w/o Post-fusion CL	0.2595	0.1502	0.0682	0.0350	0.0461	0.0350

Table 3: Component ablation results of DiGO. The complete model is highlighted in bold.

techniques such as graph-based collaborative modeling, intent disentanglement, cross-view contrastive learning, graph Transformers, and multi-view representation learning. For MultiCBR, EBRec, and EpicCBR, we rerun their publicly released implementations under the same data splits and evaluation protocol and report the average results over three independent runs. DiGO is evaluated in the same manner. For the remaining baselines, we use the results reported in their original papers.

Implementation Details. We implement DiGO in PyTorch and train it on a single NVIDIA GeForce RTX 3090

GPU. User, bundle, and item embeddings are initialized with Xavier initialization. The embedding dimension and the number of graph propagation layers are set to 64 and 2, respectively. We optimize the model using Adam with a learning rate of 10^{-3} and a batch size of 2048, and sample one negative bundle for each positive interaction. The diffusion step T is set to 9, 12, and 12 on Youshu, NetEase, and iFashion, respectively, while the maximum number of retained interactions κ is set to 7, 13, and 3. The remaining hyperparameters are tuned according to validation performance.

4.2 Overall Performance

Table 1 presents the overall performance of DiGO and the baseline methods on the three datasets.

As shown in Table 1, DiGO achieves the best performance on all evaluation metrics across the three datasets, demonstrating its effectiveness in different bundle recommendation scenarios. On Youshu, DiGO further improves both Recall and NDCG despite the strong performance of BundleGT and EpicCBR. On NetEase, DiGO consistently outperforms all baselines on the four metrics. The improvements are particularly pronounced on iFashion, where DiGO surpasses the strongest baselines by 8.23%, 11.48%, 4.95%, and 9.31% in R@10, N@10, R@20, and N@20, respectively.

Moreover, DiGO consistently outperforms MultiCBR on all three datasets. MultiCBR already exploits multi-view information through three-view graph propagation, representation fusion, and post-fusion contrastive learning. The consistent improvements achieved by DiGO demonstrate the benefit of further optimizing the user-bundle graph and aligning the two auxiliary views with the reconstructed user-bundle view before view fusion.

4.3 Ablation Study

To investigate the contribution of each component, we compare DiGO with four variants. **w/o Graph Opt.** removes diffusion-driven graph optimization and directly uses the original user–bundle graph. **w/o All CL** removes both the pre-fusion and post-fusion contrastive objectives. **w/o Pre-fusion CL** removes the cross-view contrastive learning that uses the reconstructed UB view as the anchor. **w/o Post-fusion CL** removes the self-supervised contrastive learning applied after view fusion. The results are presented in Table 3.

The contribution of graph optimization varies across datasets. Removing this component reduces R@20 and N@20 on iFashion from 0.15680 and 0.13150 to 0.14999 and 0.12446, respectively, resulting in the most noticeable degradation. It also causes performance drops on NetEase. On Youshu, R@20 remains nearly unchanged, whereas N@20 decreases. These results indicate that graph optimization provides dataset-dependent benefits and contributes most prominently on iFashion.

Removing pre-fusion cross-view contrastive learning also leads to performance degradation, with more evident changes on Youshu and NetEase. For example, R@20 and N@20 on NetEase decrease from 0.09550 and 0.05110 to 0.09057 and 0.04873, respectively. This demonstrates that aligning the UI and BI representations with the reconstructed UB view before fusion enables the auxiliary relations to more effectively supplement user–bundle supervision.

Removing post-fusion contrastive learning causes substantial performance degradation, while removing both contrastive objectives leads to a further severe decline. This observation confirms that post-fusion contrastive learning remains an important component of the multi-view recommendation backbone. Meanwhile, the pre-fusion and post-fusion contrastive objectives operate at different stages—cross-view alignment and fused-representation optimization—and jointly contribute to effective multi-view representation learning.

4.4 In-depth Analysis

Interaction Selection Strategies. To further examine whether diffusion-driven graph optimization can effectively identify reliable user–bundle interactions, we compare five graph construction strategies on iFashion. **Original UB** directly uses the observed user–bundle graph without interaction filtering. **Random- K** randomly retains at most K observed interactions for each user, while **Popularity- K** selects interactions according to the global popularity of the corresponding bundles. **Pretrain-Sim** ranks observed interactions using the cosine similarity between pretrained user and bundle representations. In contrast, **DiGO** scores each observed interaction using the overall bundle preference representation produced by conditional diffusion. All strategies select connections only from the observed interaction history, and the results are averaged over multiple independent runs.

As shown in Figure 2(a)–(b), DiGO consistently achieves the best Recall@20 and NDCG@20 under all retention

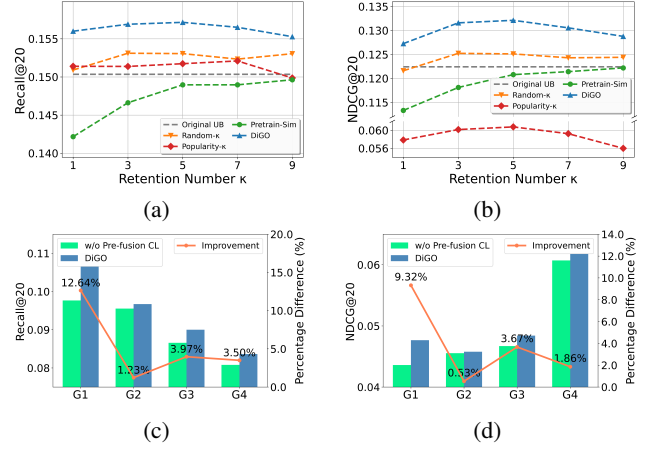


Figure 2: **In-depth analysis of DiGO.** (a)–(b) compare different interaction selection strategies on iFashion under varying retention numbers. (c)–(d) compare the performance with and without pre-fusion cross-view contrastive learning across four user groups on NetEase. Users are sorted by their training UB degrees, and G1–G4 correspond to $[0, 7)$, $[7, 8)$, $[8, 12)$, and $[12, 660]$, respectively. Graph optimization is disabled in (c)–(d), and the line reports the relative improvement introduced by pre-fusion contrastive learning.

settings. Although Random- K occasionally improves over Original UB, it remains consistently inferior to DiGO, indicating that the gains cannot be attributed merely to reducing the number of graph edges. Pretrain-Sim also performs worse than DiGO, suggesting that directly measuring the similarity between pretrained user and bundle representations is insufficient to characterize the overall preference reflected by a user’s interaction history. By modeling historical bundles through user-conditioned diffusion, DiGO provides a more effective criterion for re-evaluating observed interactions. Its performance generally first increases and then decreases as the retention number grows, reflecting the trade-off between preserving informative supervision and filtering potentially unreliable interactions.

Performance on Sparse Users. To investigate whether pre-fusion cross-view contrastive learning alleviates the limited supervision caused by sparse user–bundle feedback, we sort users according to their numbers of training UB interactions and divide them into four equally sized groups. G1 represents the sparsest users, whereas G4 represents the densest users. To isolate the effect of pre-fusion contrastive learning, diffusion-driven graph optimization is disabled in both compared settings. In Figure 2(c)–(d), **w/o Pre-fusion CL** denotes the multi-view recommendation backbone without either of the two proposed components, while **DiGO** denotes the same setting augmented only with pre-fusion cross-view contrastive learning. Therefore, the performance difference between the two variants is solely attributable to the pre-fusion alignment objective.

Introducing pre-fusion cross-view contrastive learning improves Recall@20 and NDCG@20 across all four user

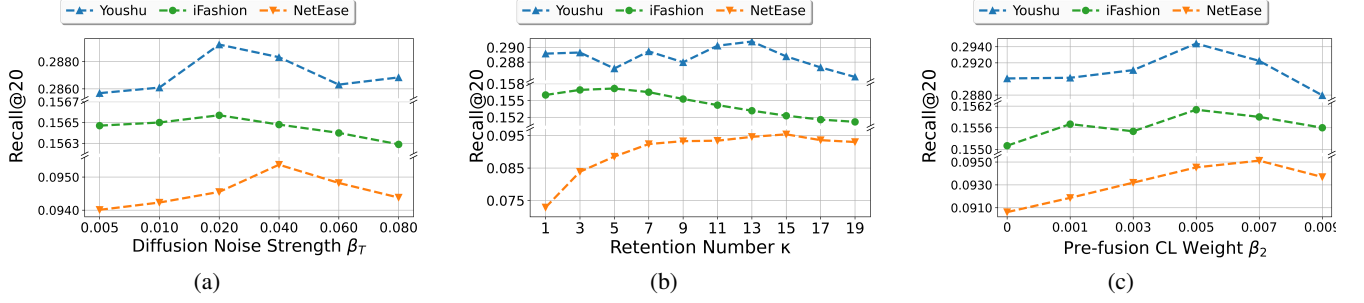


Figure 3: Hyperparameter sensitivity of DiGO on three datasets in terms of Recall@20. (a) Diffusion noise strength β_T , which denotes the maximum noise level in the diffusion schedule. (b) Retention number κ for user–bundle graph reconstruction. (c) Weight β_2 of the pre-fusion cross-view contrastive loss. Broken y -axes are used to accommodate the different performance scales across datasets.

groups. The largest gains are observed for the sparsest group G1, where Recall@20 and NDCG@20 increase by 12.64% and 9.32%, respectively. The improvements in the remaining groups are comparatively smaller, although they do not follow a strictly monotonic pattern. These results demonstrate that aligning the UI and BI representations with the UB view is particularly beneficial when direct bundle-level feedback is severely limited. By transferring complementary supervision from the two auxiliary relations before view fusion, the proposed cross-view alignment improves preference learning for sparse users.

4.5 Hyperparameter Sensitivity

We investigate the effects of the diffusion noise strength β_T , the retention number κ , and the pre-fusion contrastive loss weight β_2 , while varying one hyperparameter at a time and keeping the others fixed. The results are shown in Figure 3.

Moderate values of β_T generally yield better performance, as insufficient noise may provide limited denoising supervision, whereas excessive noise may obscure useful preference information. For κ , retaining too few interactions removes informative supervision, while retaining too many may preserve unreliable connections. Similarly, an appropriate β_2 improves cross-view alignment, whereas an overly large value may overemphasize view consistency. Overall, DiGO maintains stable performance within reasonable hyperparameter ranges.

5 Conclusion

In this paper, we proposed DiGO, a diffusion-driven graph optimization framework for bundle recommendation. To mitigate the misleading propagation caused by unreliable interactions in the user–bundle graph, DiGO models each user’s overall bundle preference through conditional diffusion and uses the resulting representation to re-evaluate observed interactions and reconstruct the user–bundle graph for subsequent multi-view learning. To address the limited supervision caused by sparse user–bundle feedback, DiGO further uses the reconstructed user–bundle view as the anchor and aligns the user–item and bundle–item views before view fusion, thereby supplementing bundle-level supervision with auxiliary relations. Experiments on three public datasets demon-

strate that DiGO outperforms existing methods across all evaluation metrics. Ablation studies, in-depth analyses, and hyperparameter experiments further validate the effectiveness and robustness of the two key designs.